

HBSC

Updated: 2026-09-06

Data Overview

It's a huge file:

```
## [1] 244097    120
```

With these columns:

HBSC	seqno_int	cluster	countryno	region
id1	id2	id3	id4	weight
adm	month	year	age	agecat
sex	grade	monthbirth	yearbirth	fasfamcar
fasbedroom	fascomputers	fasbathroom	fasdishwash	fasholidays
health	lifesat	headache	stomachache	backache
feellow	irritable	nervous	sleepdifficulty	dizzy
thinkbody	physact60	breakfastwd	breakfastwe	fruits_2
vegetables_2	sweets_2	softdrinks_2	fmeal	toothbr
timeexe	smokltm	smok30d_2	alc1tm	alc30d_2
drunkltm	drunk30d	cannabis1tm_2	cannabis30d_2	bodyweight
bodyheight	likeschool	schoolpressure	studtogether	studhelpful
studaccept	teacheraccept	teachercare	teachertrust	bulliedothers
beenbullied	cbulliedothers	cbeenbullied	fight12m	injured12m
friendhelp	friendcounton	friendshare	friendtalk	emconlfreq1
emconlfreq2	emconlfreq3	emconlfreq4	emconlpref1	emconlpref2
emconlpref3	emcsocmed1	emcsocmed2	emcsocmed3	emcsocmed4
emcsocmed5	emcsocmed6	emcsocmed7	emcsocmed8	emcsocmed9
hadsex	agesex	contraceptcondom	contraceptpill	countryborn
countrybornmo	countrybornfa	motherhome1	fatherhome1	stepmohome1
stepfahome1	fosterhome1	elsehome1_2	employfa	employmo
employnotfa	employnotmo	talkfather	talkstepfa	talkmother
talkstepmo	famhelp	famsup	famtalk	famdec
MBMI	IRFAS	IRRELFAS_LMH	IOTF4	oweight_who

So after selecting and grouping variables, create dat with :

```
## ## Individual Information
## * 'seqno_int'
## * 'countryno'
## * 'agecat'
## * 'sex'
## * 'MBMI'
## * 'IRFAS'
```

```

## * 'IRRELFAS_LMH'
## * 'IOTF4'
## * 'bodyweight'
## * 'bodyheight'
##
## ## Health Issues and Life Satisfaction
## * 'health'
## * 'lifesat'
## * 'headache'
## * 'stomachache'
## * 'backache'
## * 'feellow'
## * 'irritable'
## * 'nervous'
## * 'sleepdifficulty'
## * 'dizzy'
##
## ## Problematic Social Media Use (PMSU)
## * 'emcsocmed1'
## * 'emcsocmed2'
## * 'emcsocmed3'
## * 'emcsocmed4'
## * 'emcsocmed5'
## * 'emcsocmed6'
## * 'emcsocmed7'
## * 'emcsocmed8'
## * 'emcsocmed9'
##
## ## Bullying
## * 'beenbullied'
## * 'cbeenbullied'
##
## ## Communicates Online More Than Real Life
## * 'emconlfreq1'
## * 'emconlfreq2'
## * 'emconlfreq3'
## * 'emconlfreq4'
## * 'emconlpref1'
## * 'emconlpref2'
## * 'emconlpref3'
##
## ## Family Support
## * 'famhelp'
## * 'famsup'
## * 'famtalk'
## * 'famdec'
##
## ## Friends Support
## * 'friendhelp'
## * 'friendcounton'
## * 'friendshare'
## * 'friendtalk'
##
## ## School Support

```

```

## * 'likeschool'
## * 'schoolpressure'
## * 'studtogether'
## * 'studhelpful'
## * 'studaccept'
## * 'teacheraccept'
## * 'teachercare'
## * 'teachertrust'
##
## ## Communication with Parents
## * 'talkfather'
## * 'talkmother'
##
## ## Good/Bad Habits
## * 'physact60'
## * 'timeexe'
## * 'alc30d_2'

```

The number of missing values is relevant and need to address later:

##	variable	na_count	na_pct
## 1	IOTF4	54541	22%
## 2	MBMI	53311	22%
## 3	bodyweight	41128	17%
## 4	bodyheight	40036	16%
## 5	emcsocmed8	32878	13%
## 6	emcsocmed9	32855	13%
## 7	emcsocmed7	32597	13%
## 8	emcsocmed6	32544	13%
## 9	emcsocmed5	32443	13%
## 10	emcsocmed4	32428	13%
## 11	emcsocmed3	32009	13%
## 12	emcsocmed1	31973	13%
## 13	emcsocmed2	31659	13%
## 14	emconlpref3	31097	13%
## 15	emconlpref2	30586	13%
## 16	emconlpref1	29813	12%
## 17	emconlfreq2	29552	12%
## 18	emconlfreq4	29003	12%
## 19	emconlfreq3	28973	12%
## 20	emconlfreq1	26720	11%
## 21	famdec	20237	8%
## 22	famsup	20091	8%
## 23	famtalk	20019	8%
## 24	alc30d_2	19326	8%
## 25	famhelp	18896	8%
## 26	teachercare	18340	8%
## 27	teachertrust	17839	7%
## 28	talkfather	17491	7%
## 29	studaccept	17444	7%
## 30	beenbullied	16962	7%
## 31	teacheraccept	16764	7%
## 32	studhelpful	16746	7%
## 33	studtogether	16351	7%

```
## 34      talkmother      15521      6%
## 35      cbeenbullied    15084      6%
## 36          IRFAS      12599      5%
## 37      IRRELFAS_LMH    12599      5%
## 38      sleepdifficulty  10814      4%
## 39      friendshare     10813      4%
## 40      friendtalk      10808      4%
## 41      friendcounton    10374      4%
## 42      friendhelp       9744      4%
## 43          dizzy       8566      4%
## 44          fellow       6696      3%
## 45          backache     6543      3%
## 46          irritable     6539      3%
## 47          nervous      6525      3%
## 48      physact60        6215      3%
## 49      stomachache      5679      2%
## 50      schoolpressure   5486      2%
## 51      likeschool       5109      2%
## 52          timeexe      4799      2%
## 53          lifesat       4512      2%
## 54          headache     4505      2%
## 55          health       2577      1%
## 56          agecat       1516      1%
## 57      seqno_int         0      0%
## 58      countryno         0      0%
## 59          sex          0      0%
```

Now I explore all variables by group and feature engineer as appropriate.

[1] “Individual Information”

11 vars here. So do 1x1.

```
## [1] "seqno_int"      "countryno"      "agecat"         "sex"            "MBMI"
## [6] "IRFAS"          "IRRELFAS_LMH"  "IOTF4"          "bodyweight"     "bodyheight"
```

seqno_int is just an identifier.

countryno is recoded to country_name and this is the distribution.

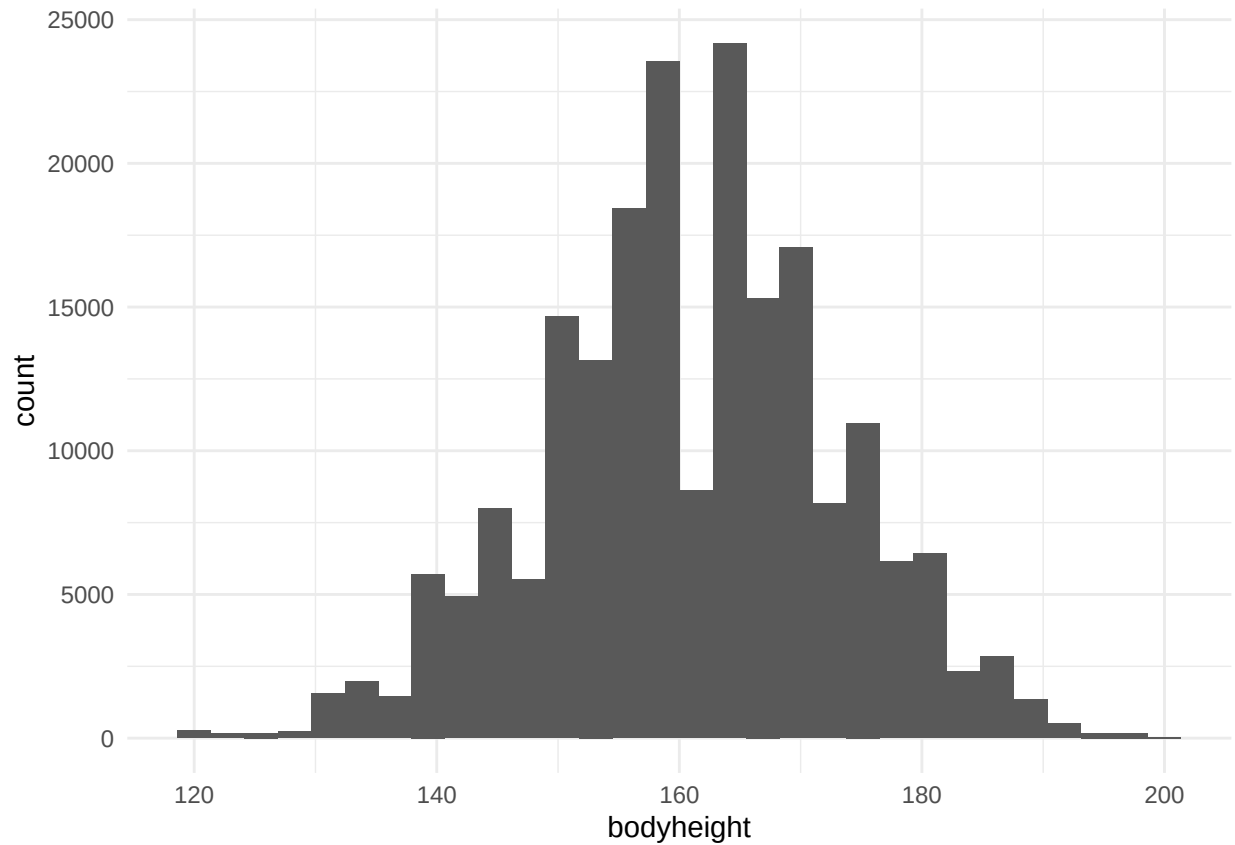
Table 2: #/% of observations by country

country_name	n	pct
Albania	1765	1%
Armenia	4717	2%
Austria	4129	2%
Azerbaijan	4586	2%
Belgium (Flemish)	4333	2%
Belgium (French)	5578	2%
Canada	12950	5%
Croatia	5169	2%

country_name	n	pct
Czech Republic	11564	5%
Denmark	3181	1%
England	3397	1%
Estonia	4725	2%
Finland	3146	1%
France	9170	4%
Georgia	4242	2%
Germany	4347	2%
Greenland	1243	1%
Hungary	3789	2%
Iceland	6996	3%
Ireland	3833	2%
Israel	7712	3%
Italy	4144	2%
Kazakhstan	4868	2%
Latvia	4412	2%
Lithuania	3797	2%
Luxembourg	4070	2%
Macedonia	4658	2%
Malta	2576	1%
Netherlands	4698	2%
Norway	3127	1%
Poland	5224	2%
Portugal	6126	3%
Republic of Moldova	4686	2%
Romania	4567	2%
Russia	4281	2%
Scotland	5021	2%
Serbia	3933	2%
Slovakia	4785	2%
Slovenia	5667	2%
Spain	4320	2%
Sweden	4185	2%
Switzerland	7510	3%
Turkey	5848	2%
Ukraine	6660	3%
Wales	15951	7%
NA	8411	3%

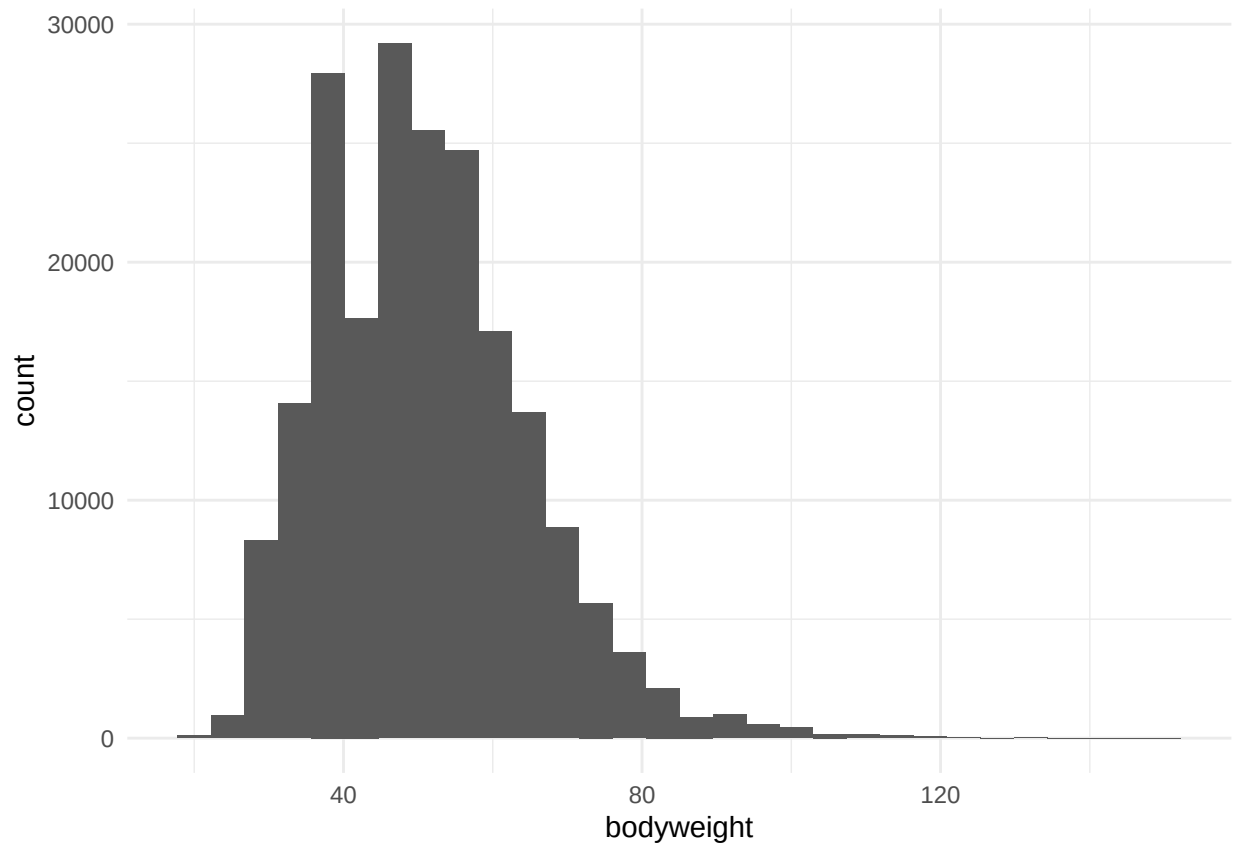
bodyheight is in cm and range from 120 to 200 so will use. Impute NAs?

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	120.0	153.0	161.0	161.1	170.0	200.0	40036



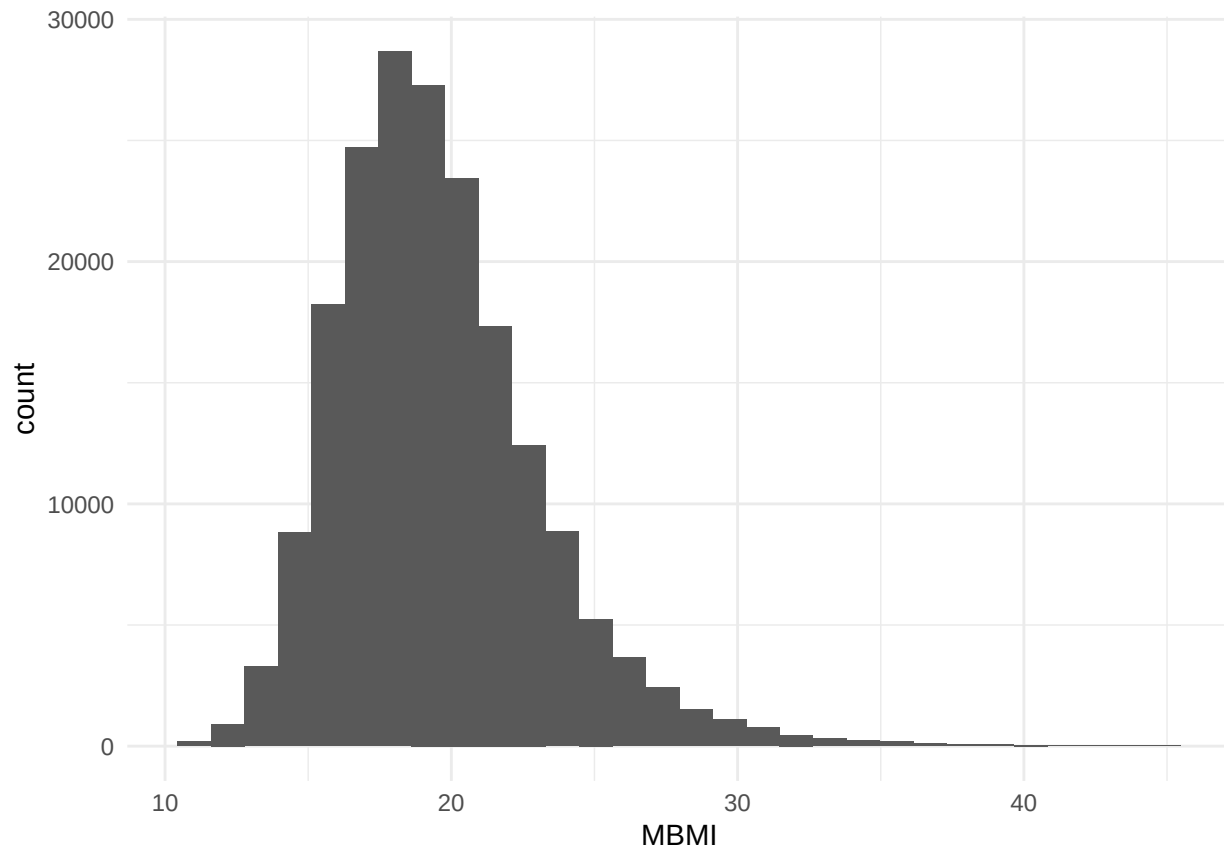
bodyweight is in kilo and range from 20 to 150 so will use maybe. Impute NAs?

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	20.00	40.00	50.00	51.07	60.00	150.00	41128



MBMI is body mass index from [Range= 11.02-44.9] and 0 meaning outside overall range so make those NA.

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	0.00	17.09	19.04	19.51	21.40	44.91	53311



IOTF4 is a BMI group from 1(thin) to 4(observe) so leave as is.

```
##
##      1      2      3      4  <NA>
## 26967 132661 24487  5441 54541
```

agecat has 3 values and will leave as is: 1 ~ 11, 2 ~ 13, 3 ~ 15.

```
##
##      1      2      3  <NA>
## 81834 83749 76998 1516
```

sex is binary with 1 (boy), 2 (girl) so will leave as is.

```
##
##      1      2  <NA>
## 120350 123747    0
```

IRRELFAS_LMH is relative family affluence with 1 Low20, 2 Mid60, 3 High20. Leave as is.

IRFAS is discrete [0,13] and shows affluence.

```
##
##      1      2      3  <NA>
##    0    715    0    0    0
```

```
## 1 2238 2 0 0
## 2 4780 270 0 0
## 3 5217 3609 0 0
## 4 6829 6272 0 0
## 5 8199 9442 0 0
## 6 8522 13305 331 0
## 7 7593 17420 1513 0
## 8 1724 26873 2120 0
## 9 1 29479 2672 0
## 10 0 23830 5408 0
## 11 0 11158 11359 0
## 12 0 488 13893 0
## 13 0 0 6236 0
## <NA> 0 0 0 12599
```

[2] “Health Issues and Life Satisfaction”

10 vars here:

```
## [1] "health" "lifesat" "headache" "stomachache"
## [5] "backache" "feellow" "irritable" "nervous"
## [9] "sleepdifficulty" "dizzy"
```

health is self rating of health from 1(Excellent) to 4(poor). Won't use for now

```
table(dat$health, useNA = "always")
```

```
##
## 1 2 3 4 <NA>
## 86985 120155 30473 3907 2577
```

lifesat asks for life satisfaction as ladder where 0(bottom) to 10(best). I'll dichotomize and keep num for shap.

lifesat distribution:

```
##
## 0 1 2 3 4 5 6 7 8 9 10 <NA>
## 1218 1041 1748 3691 6961 17137 19921 39120 55325 45271 48152 4512
```

lifesat %:

```
##
## 0 1 2 3 4 5 6 7 8 9 10 <NA>
## 0.005 0.004 0.007 0.015 0.029 0.070 0.082 0.160 0.227 0.185 0.197 0.018
```

lifesat_low as y/n:

```
##
##          0      1  <NA>
##  0          0 1218      0
##  1          0 1041      0
##  2          0 1748      0
##  3          0 3691      0
##  4          0 6961      0
##  5          0 17137     0
##  6    19921      0      0
##  7    39120      0      0
##  8    55325      0      0
##  9    45271      0      0
## 10    48152      0      0
##  <NA>      0      0  4512
```

headache, stomachache, backache, and dizzy are physical and will ignore for now. But many papers report on psychosomatic issues so I might come back.

feellow, irritable, nervous, sleepdifficulty and mental health issues. They answer a question In the last 6 months: how often have you had the following...? And the answers goes from 1 (About every day) to 5 (Rarely or never). Their distribution is:

```
## $feellow
##
##      1      2      3      4      5
## 20224 25739 29784 48898 112756
##
## $irritable
##
##      1      2      3      4      5
## 26053 35879 42648 57922 75056
##
## $nervous
##
##      1      2      3      4      5
## 28306 32770 40064 54109 82323
##
## $sleepdifficulty
##
##      1      2      3      4      5
## 30501 25543 26053 36570 114616
```

Their correlation is modest.

```
##          feellow irritable nervous sleepdifficulty
## feellow          1.00      0.54      0.48          0.36
## irritable          0.54      1.00      0.53          0.35
## nervous           0.48      0.53      1.00          0.36
## sleepdifficulty    0.36      0.35      0.36          1.00
```

These variables are then recoded to a binary where it flags if the complaint occurs more than once per week (values 1/2).

These frequency 1/0 vars are summed up and if sum is 2+ then there's an issue. And counts look like this:

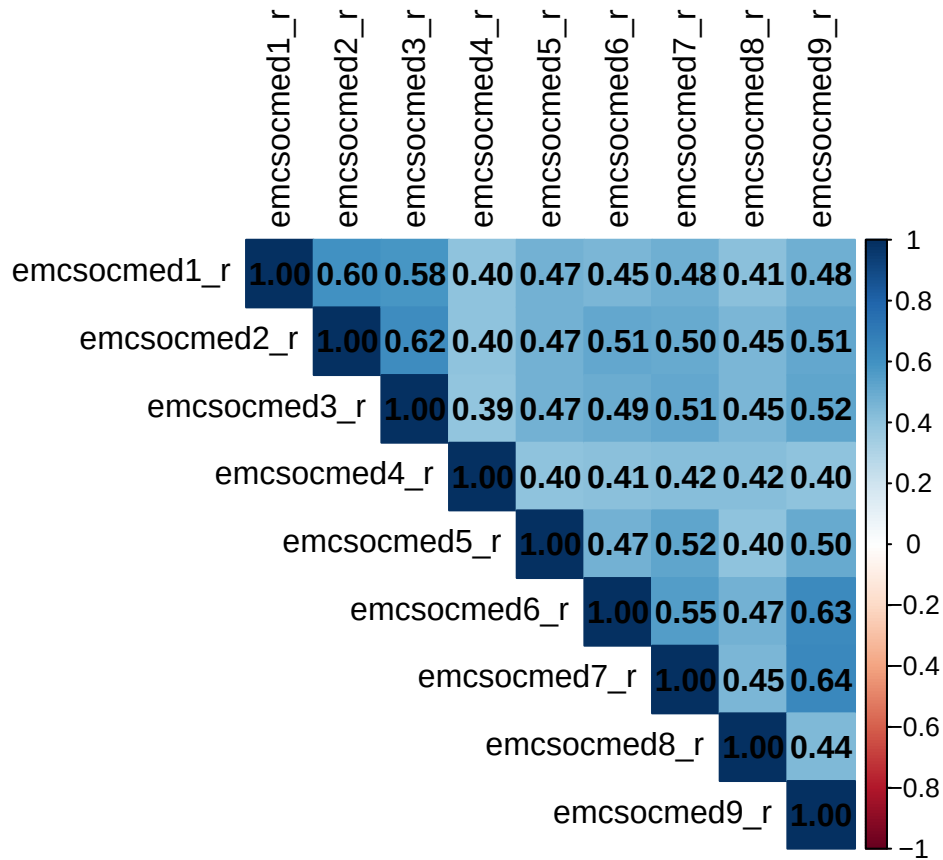
```
##
##           0           1   <NA>
##    0    121573         0       0
##    1     45509         0       0
##    2         0    29991       0
##    3         0    21172       0
##    4         0    12970       0
##   <NA>         0         0  12882
```

[3] “Problematic Social Media Use (PMSU)”

9 binary variables are recoded to 0/1.

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

And they are correlated.



These 9 vars are sumed and then binned to H/M/L where H(6-9), M(2-5) and L(0-1). Resulting in:

	1	2	3	NA
0	72474	0	0	0
1	36277	0	0	0
2	0	26066	0	0
3	0	19012	0	0
4	0	14908	0	0
5	0	10306	0	0
6	0	0	5905	0
7	0	0	3382	0
8	0	0	1878	0
9	0	0	2979	0
NA	0	0	0	50910

[4] “Bullying”

There are 2 variables that measure if kid has been bullied. a) beenbullied: Been bullied past months and b) cbeenbullied: Been cyber bullied. Answers range from 1 (Haven’t) to 5 (Several times a week).

I will derive 2 to keep: cbeenbullied as y/n and bulliedsum as sum of both.

```
## cbullied recoding:
```

```
##
##           0      1  <NA>
##  1  198900      0      0
##  2       0  20426      0
##  3       0   4625      0
##  4       0   2274      0
##  5       0   2788      0
##  <NA>      0      0 15084
```

bullying score: (low is better)

score	n	pct
2	151013	62%
3	34034	14%
4	13753	6%
5	7360	3%
6	7703	3%
7	2745	1%
8	1288	1%
9	747	0%
10	1093	0%
NA	24361	10%

[5] “Communicates Online More Than Real Life”

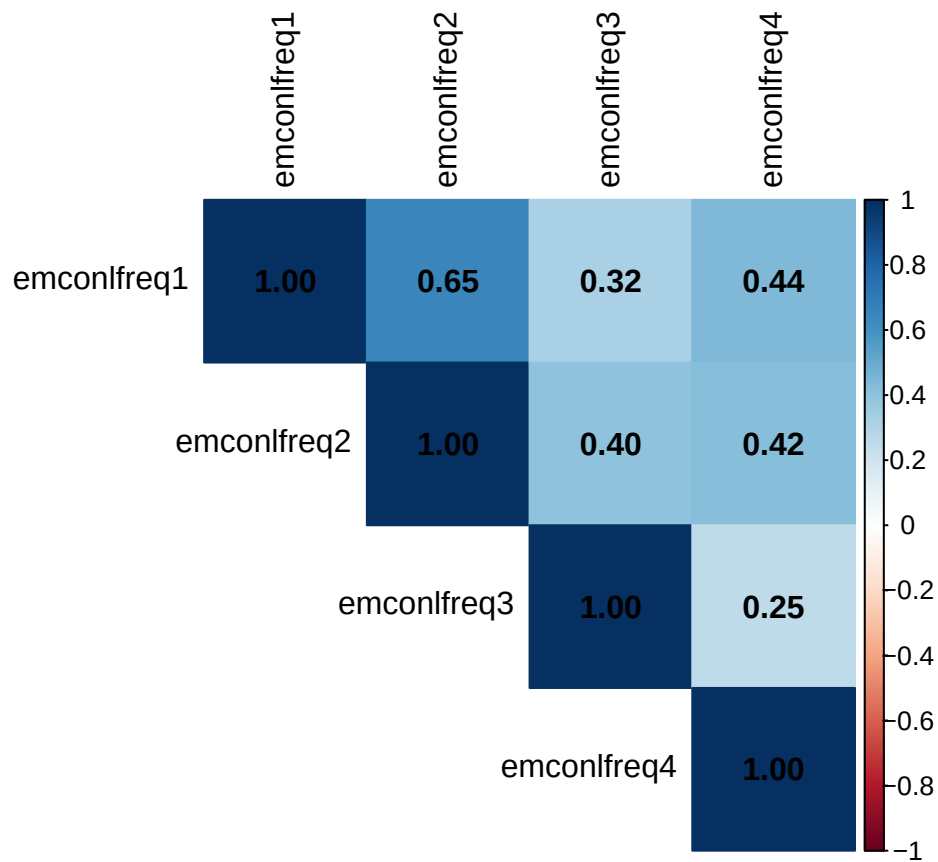
7 vars here in 2 themes - a) continuous online communications and b) preference to communicate online.

For a) continuous online communications, measures how often young person communicate online with 4 types of people in their life. Answers range from 1 (NA or don't know) to 6 (almost all of the time).

Variable	Description
emconlfreq1	Online contact with close friends
emconlfreq2	Online contact with larger friend group
emconlfreq3	Online contact with online friends
emconlfreq4	Online contact with other

And they are somewhat correlated but not much.

```
## [1] "emconlfreq1" "emconlfreq2" "emconlfreq3" "emconlfreq4" "emconlpref1"
## [6] "emconlpref2" "emconlpref3"
```



I combine them into two options. a) one dichotomized and adopt the HBSC definition of yes if any of the variables is 6; and b) a sum for our shap tests with values in [4,24].

```
## Online frequency is (onlfreq_is):
```

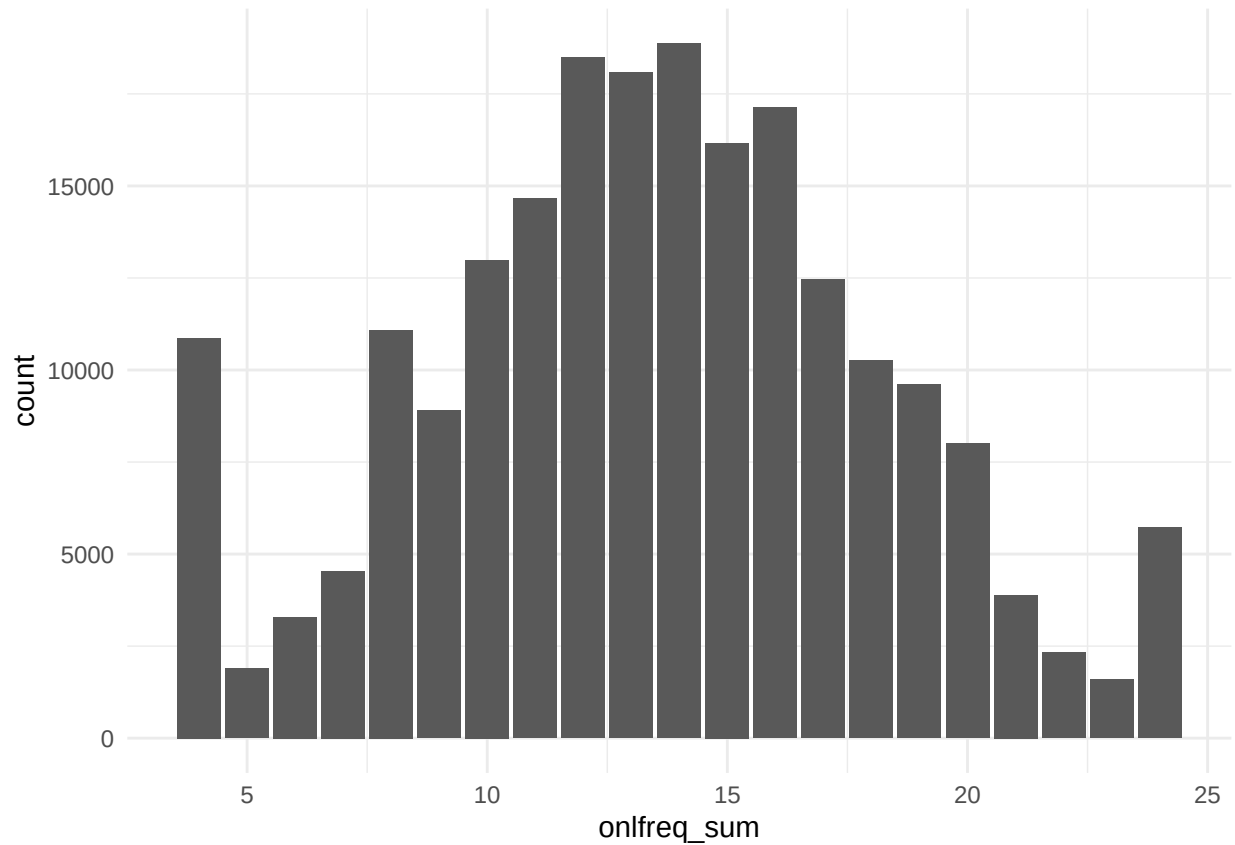
```
##
```

```
##      0      1  <NA>
```

```
## 143585 74898 25614
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
```

```
##   4.00  10.00   14.00   13.51  17.00   24.00  33251
```

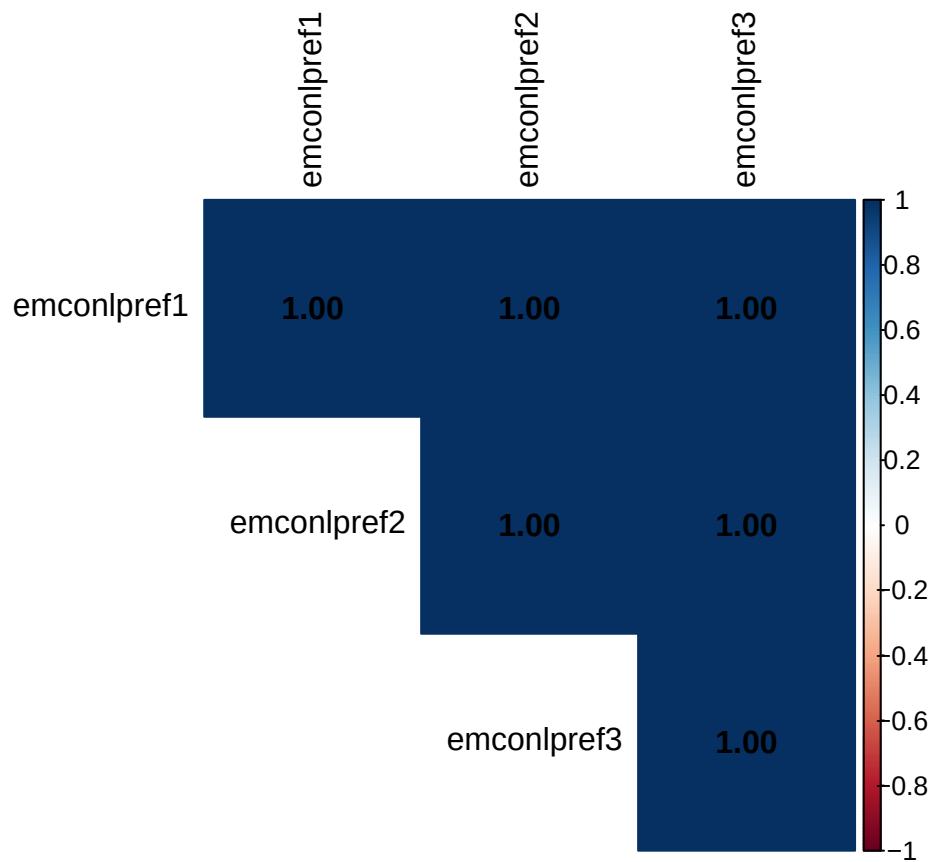


For b) preference to communicate online, 3 questions ask about preference of sharing important stuff online vs real-life. The three variables are:

Variable	Description
emconlpref1	Secrets, more easily online
emconlpref2	Feelings, more easily online
emconlpref3	Concerns, more easily online

The answers range from 1 (Strong disagree) to 5 (Strong agree).

And they are correlated. Remove 99s????

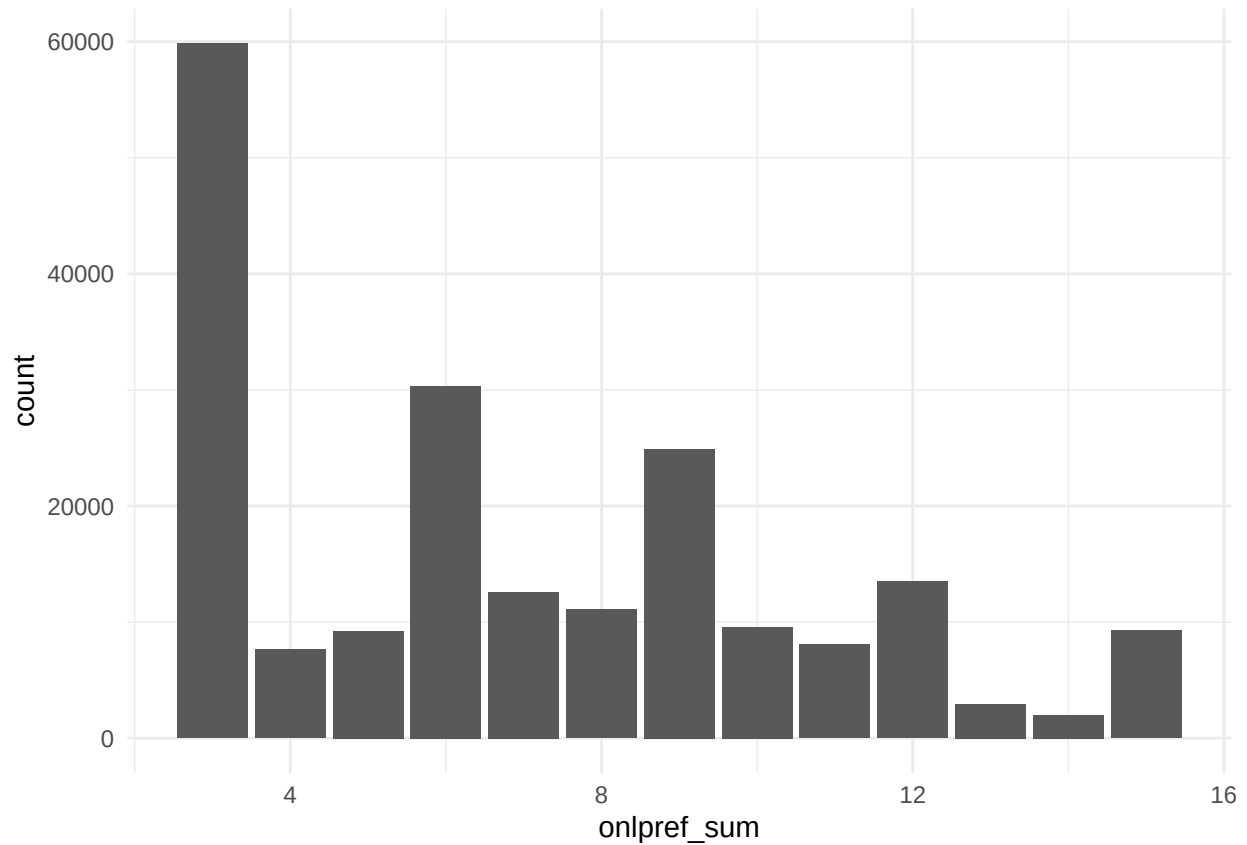


Haven't found a paper that uses them so I will replicated method of friends support below. Summing them results in [3,15] and then grouped as H/M/L as L 3-6, M 7-10, H 11-15. Resulting in:

```
## onlpref:
```

```
##
##      1      2      3  <NA>
## 107085 58134 35713 43165
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      3.000   3.000   6.000   6.919   9.000  15.000  43165
```



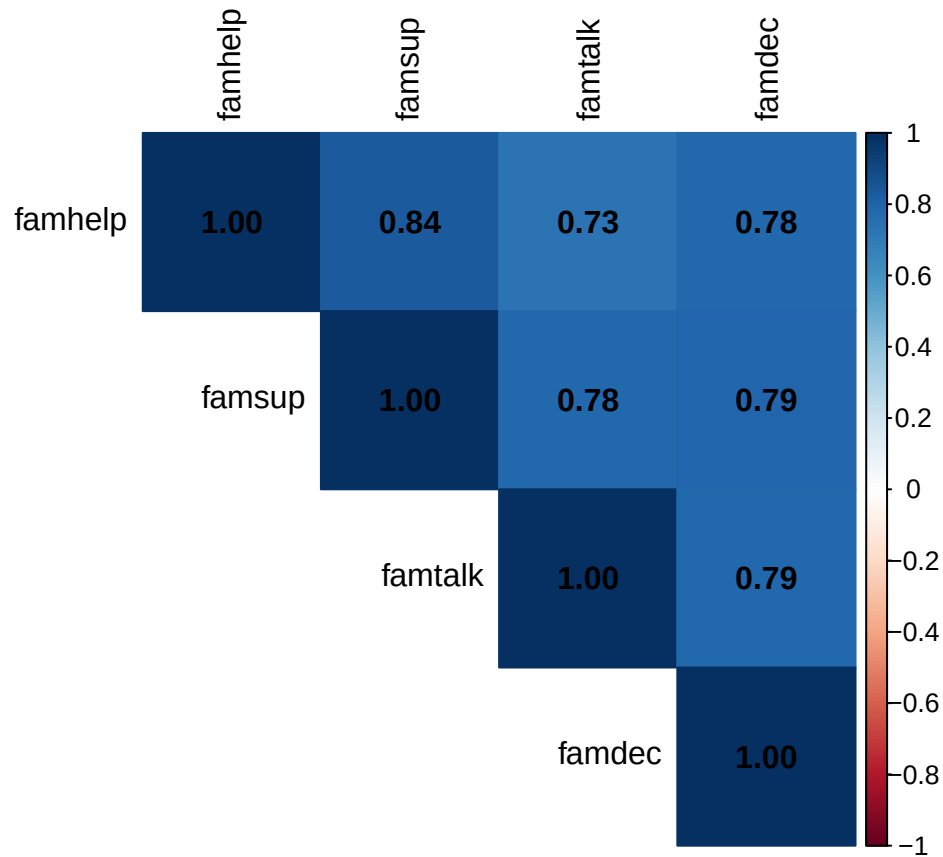
[6] “Family Support”

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "famhelp" "famsup" "famtalk" "famdec"
```

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family
famclose	Feel close to family

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	10939	0	0	0
5	1597	0	0	0
6	1452	0	0	0
7	1382	0	0	0
8	2477	0	0	0
9	1485	0	0	0
10	2274	0	0	0
11	1739	0	0	0
12	0	2655	0	0
13	0	2279	0	0
14	0	2745	0	0
15	0	2781	0	0
16	0	5747	0	0
17	0	3440	0	0
18	0	4455	0	0
19	0	4637	0	0
20	0	0	7739	0
21	0	0	5853	0
22	0	0	9240	0
23	0	0	7878	0
24	0	0	13474	0
25	0	0	11089	0

	1	2	3	NA
26	0	0	13090	0
27	0	0	15938	0
28	0	0	85359	0
NA	0	0	0	22353

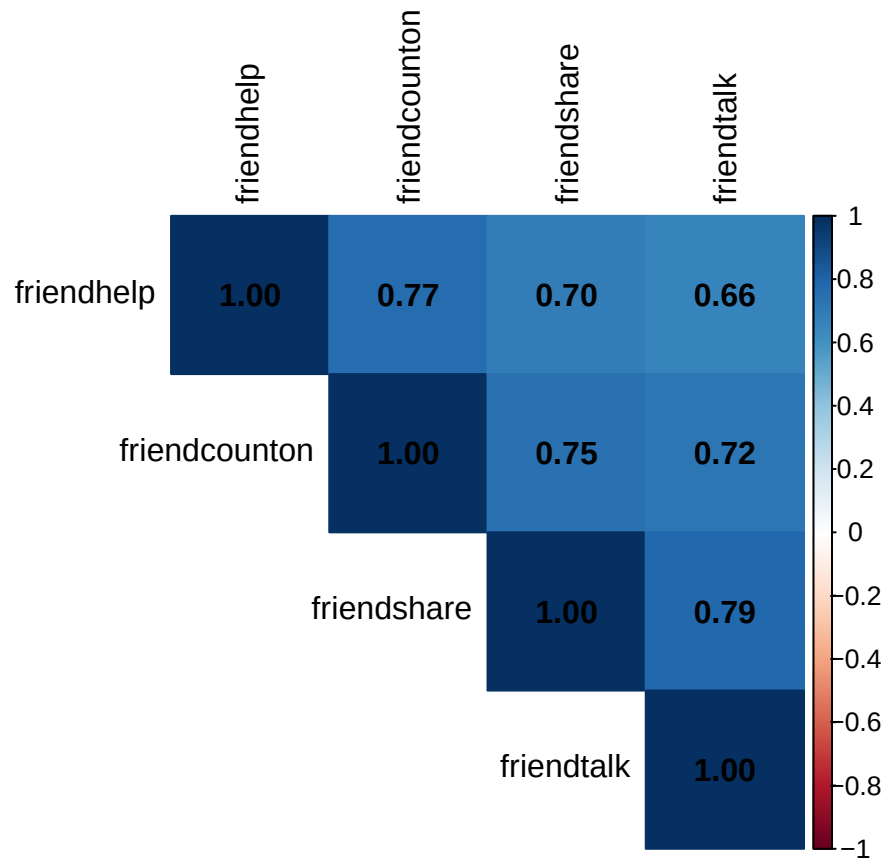
[7] “Friends Support”

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "friendhelp"      "friendcounton" "friendshare"    "friendtalk"
```

item	topic
friendhelp	My friends really try to help me
friendcounton	I can count on my friends when things go wrong
friendshare	I have friends with whom I can share my joys and sorrows
friendtalk	I can talk about my problems with my friends

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	9995	0	0	0
5	2159	0	0	0
6	2109	0	0	0
7	2392	0	0	0
8	3798	0	0	0
9	2463	0	0	0
10	3752	0	0	0
11	2869	0	0	0
12	0	4267	0	0
13	0	3834	0	0
14	0	4428	0	0
15	0	4339	0	0
16	0	9075	0	0
17	0	5750	0	0
18	0	7126	0	0
19	0	7644	0	0
20	0	0	11463	0
21	0	0	8358	0
22	0	0	12583	0
23	0	0	10381	0
24	0	0	16855	0
25	0	0	12468	0
26	0	0	12977	0
27	0	0	14317	0
28	0	0	55799	0
NA	0	0	0	12896

[8] “School Support”

8 variables related to school.

```
## $'School Support'
## [1] "likeschool"      "schoolpressure" "studtogether"   "studhelpful"
## [5] "studaccept"      "teacheraccept"  "teachercare"    "teachertrust"
```

likeschool ask if they like school and goes from 1 (lots) to 4 (no). I won't use for now.

schoolpressure ask if they feel pressure and goes from 1(no) to 4(a lot). I won't use for now.

Then there are 3 questions about relationship with students and 3 with relationship with teachers. The answers go from 1 (strong agree) to 5 (strong disagree).

category	item	topic
teacher	teacheraccept	Teacher accepts me
teacher	teachercare	Teacher cares about me
teacher	teachertrust	Feel trust in teacher
classmates	studtogether	Students enjoy being together
classmates	studhelpful	Students kind and helpful
classmates	studaccept	Students accept me

Scales are reversed so 1 is the negative and 5 is the positive.

And there is moderate correlation between groups but lowers across stud/teach.

```
##          teacheraccept_r teachercare_r teachertrust_r studtogether_r
## teacheraccept_r          1.00          0.60          0.59          0.28
## teachercare_r            0.60          1.00          0.65          0.27
## teachertrust_r           0.59          0.65          1.00          0.28
## studtogether_r           0.28          0.27          0.28          1.00
## studhelpful_r            0.30          0.30          0.32          0.56
## studaccept_r             0.35          0.27          0.27          0.45
##          studhelpful_r studaccept_r
## teacheraccept_r          0.30          0.35
## teachercare_r            0.30          0.27
## teachertrust_r           0.32          0.27
## studtogether_r           0.56          0.45
## studhelpful_r            1.00          0.54
## studaccept_r             0.54          1.00
```

A few features are derived. For stud/teach separate a binary following HBSC and a sum for shap test.

teachersup:

```
##
##          0      1  <NA>
## 3      3593      0      0
## 4      1483      0      0
## 5      2657      0      0
## 6      5531      0      0
## 7      6727      0      0
## 8      9789      0      0
## 9     19836      0      0
## 10     22972      0      0
## 11     27699      0      0
## 12         0 40136      0
## 13         0 23751      0
## 14         0 21349      0
## 15         0 38813      0
## <NA>         0      0 19761
```

studentsup:

```
##
##          0      1  <NA>
## 3      2138      0      0
## 4      1199      0      0
## 5      2018      0      0
## 6      4392      0      0
## 7      6367      0      0
## 8      9336      0      0
## 9     18744      0      0
## 10     22280      0      0
## 11     30233      0      0
```

```
##      12      0 48125      0
##      13      0 26851      0
##      14      0 21371      0
##      15      0 32245      0
##    <NA>      0      0 18798
```

[9] “Communication with Parents”

Two vars with answers from 1 (Very easy) to 5 (Don’t have or see). With this counts and proportions in data.

```
## [1] "talkfather" "talkmother"
```

```
##           talkmother
## talkfather      1      2      3      4      5 <NA>
##           1  62315  6269  2345   842   891 1115
##           2  31505 40099  4150   959   753   880
##           3   9936 17764 10530  1130   499   430
##           4   3719  5070  4602  5132   294   193
##           5   5842  4188  1968  1039  1905   242
##          <NA>  2582  1523   447   174   104 12661
```

```
##           talkmother
## talkfather      1      2      3      4      5 <NA>
##           1  0.255 0.026 0.010 0.003 0.004 0.005
##           2  0.129 0.164 0.017 0.004 0.003 0.004
##           3  0.041 0.073 0.043 0.005 0.002 0.002
##           4  0.015 0.021 0.019 0.021 0.001 0.001
##           5  0.024 0.017 0.008 0.004 0.008 0.001
##          <NA> 0.011 0.006 0.002 0.001 0.000 0.052
```

With some correlation between them (I expected more).

So I will dichotomize like HBSC and keep sum of both as well for SHAP. These are dichotomized for each parent as Yes if [1,2] and No if [3-5].

Where new vars look like this for dichotom values:

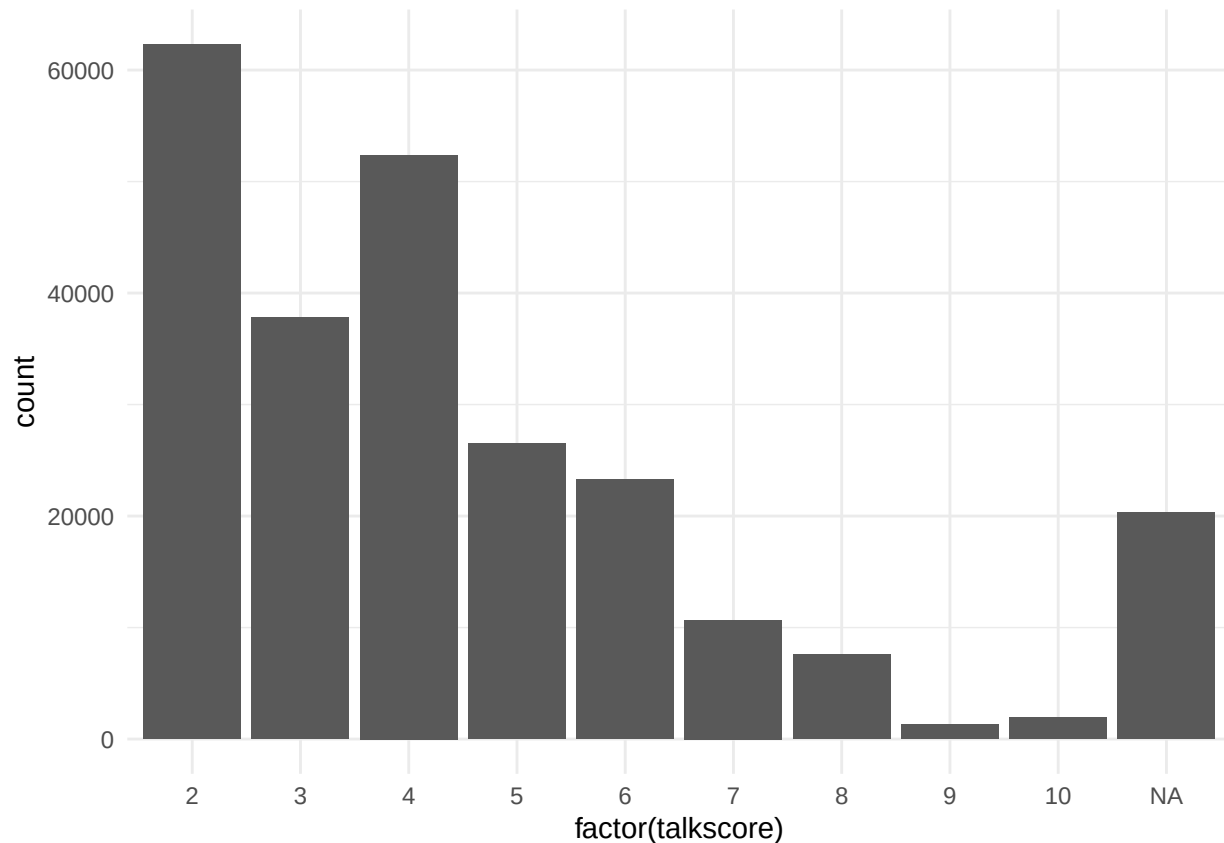
```
## Father:
```

```
##
##           0      1 <NA>
##      1      0 73777      0
##      2      0 78346      0
##      3  40289      0      0
##      4  19010      0      0
##      5  15184      0      0
##    <NA>      0      0 17491
```

```
## Mother:
```

```
##
##           0      1  <NA>
##    1         0 73777      0
##    2         0 78346      0
##    3    40289      0      0
##    4    19010      0      0
##    5    15184      0      0
##   <NA>      0      0 17491
```

And the talk score looks like this. Will keep for now and orig vars as well.



[10] “Good/Bad Habits”

physact60 is just past week so ignore.

timeexe ask how often exercise in free time going from 1 (every day) to 7 never. HBSC shows proportions of those that do 3+ per week. But that is 2022 data, in our case we do 2-3 times per week or more. And I keep timeexe as is for shap test.

```
##
##           0      1  <NA>
##    1         0 41797      0
##    2         0 56673      0
##    3         0 71661      0
##    4    32246      0      0
```

```
##    5    10541    0    0
##    6    10256    0    0
##    7    16124    0    0
##   <NA>      0    0  4799
```

alc30d_2 ask about alcohol use in past 30 days and will ignore for now

Clean Dataset for Modeling

Missing Values

Need to plan for this. For now, just keeping complete obs, but note some countries are very affected.

country_name	n	na_FALSE	pct_complete
Portugal	6126	4955	0.809
Finland	3146	2351	0.747
Estonia	4725	3470	0.734
Hungary	3789	2779	0.733
Italy	4144	3007	0.726
Lithuania	3797	2721	0.717
Austria	4129	2923	0.708
Slovenia	5667	3974	0.701
Republic of Moldova	4686	3230	0.689
Spain	4320	2964	0.686
Czech Republic	11564	7877	0.681
Serbia	3933	2602	0.662
Belgium (Flemish)	4333	2836	0.655
Ukraine	6660	4341	0.652
Netherlands	4698	3058	0.651
Germany	4347	2824	0.650
Iceland	6996	4430	0.633
Russia	4281	2604	0.608
Albania	1765	1056	0.598
Turkey	5848	3466	0.593
Croatia	5169	2934	0.568
Sweden	4185	2353	0.562
France	9170	5056	0.551
Romania	4567	2500	0.547
Luxembourg	4070	2171	0.533
Israel	7712	4027	0.522
Georgia	4242	2127	0.501
Norway	3127	1472	0.471
Kazakhstan	4868	2179	0.448
Canada	12950	5591	0.432
Armenia	4717	1950	0.413
Belgium (French)	5578	2207	0.396
Malta	2576	612	0.238
Scotland	5021	1068	0.213
Wales	15951	2832	0.178
Ireland	3833	603	0.157
England	3397	254	0.075

country_name	n	na_FALSE	pct_complete
Azerbaijan	4586	0	0.000
Denmark	3181	0	0.000
Greenland	1243	0	0.000
Latvia	4412	0	0.000
Macedonia	4658	0	0.000
Poland	5224	0	0.000
Slovakia	4785	0	0.000
Switzerland	7510	0	0.000
NA	8411	0	0.000

Data Type Coercions

```

# # vars to coerce to fact (i looked 1x1)
# country_name
# IOTF4 ordered 1-4
# agecat ordered 1-3
# sex
# IRRELFAS_LMH ordered 1-3
# lifesat_low ordered 0-1
# mental_issue ordered 0-1
# pmsu_lmh order 1-3
# cbullied order 0-1
# onlfreq_is order 0-1
# onlpref order 1-3
# famsup_hml order 1-3
# frisup_hml order 1-3
# teachersup order 0-1
# studsup order 0-1
# talkf order 0-1
# talkm order 0-1
# timeexe_r order 0-1

d <- d %>%
  mutate(
    # --- Unordered (Nominal) Factors ---
    country_name = as.factor(country_name),
    sex = as.factor(sex),
    # --- Ordered Factors (1 to 4) ---
    IOTF4 = factor(IOTF4, levels = c("1", "2", "3", "4"), ordered = TRUE),
    # --- Ordered Factors (1 to 3) ---
    agecat = factor(agecat, levels = c("1", "2", "3"), ordered = TRUE),
    IRRELFAS_LMH = factor(IRRELFAS_LMH, levels = c("1", "2", "3"), ordered = TRUE),
    pmsu_lmh = factor(pmsu_lmh, levels = c("1", "2", "3"), ordered = TRUE),
    onlpref = factor(onlpref, levels = c("1", "2", "3"), ordered = TRUE),
    famsup_hml = factor(famsup_hml, levels = c("1", "2", "3"), ordered = TRUE),
    frisup_hml = factor(frisup_hml, levels = c("1", "2", "3"), ordered = TRUE),
    # --- Ordered Factors (0 to 1 Binary / Dummy Scales) ---
    lifesat_low = factor(lifesat_low, levels = c("0", "1"), ordered = TRUE),
    mental_issue = factor(mental_issue, levels = c("0", "1"), ordered = TRUE),
  )

```

```

cbullied      = factor(cbullied,      levels = c("0", "1"), ordered = TRUE),
onlfreq_is    = factor(onlfreq_is,    levels = c("0", "1"), ordered = TRUE),
teachersup    = factor(teachersup,    levels = c("0", "1"), ordered = TRUE),
studsup       = factor(studsup,       levels = c("0", "1"), ordered = TRUE),
talkf         = factor(talkf,         levels = c("0", "1"), ordered = TRUE),
talkm         = factor(talkm,         levels = c("0", "1"), ordered = TRUE),
timeexe_r     = factor(timeexe_r,     levels = c("0", "1"), ordered = TRUE)
)

```

Tests for Potential of Prediction

```

#vars_for_cochran (ordered)
#vars_for_chi
#vars_for_tt

# --- 1. SETUP YOUR TARGET VARIABLE NAME ---
# Change "depression" to your exact binary response column name
# IMPORTANT: This target variable must have exactly two levels (e.g., "No" and "Yes")
#target_var <- "depression"
target_var <- "mental_issue"
# Ensure target is a factor and identify the "success" level (the 2nd level)
#d[[target_var]] <- as.factor(d[[target_var]])
target_levels <- levels(d[[target_var]])
success_level <- target_levels[2]

# --- 2. THE AUTOMATED LOOP ---
for (col_name in names(d)) {

  # Skip the target variable itself
  if (col_name %in% c(target_var, "seqno_int")) next

  cat("\n===== \n")
  cat("Testing Relationship:", target_var, "vs", col_name, "\n")
  cat("===== \n")

  # --- 1. IF ORDINAL -> RUN COCHRAN-ARMITAGE TREND TEST ---
  if (is.ordered(d[[col_name]])) {
    cat("Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...\n\n")

    # Build table: rows = target, columns = ordinal levels
    contingency_table <- table(d[[target_var]], d[[col_name]])
    print(contingency_table)
    cat("\n")

    # Extract successes for the 2nd level (e.g., "Yes") and column totals
    successes <- contingency_table[success_level, ]
    totals    <- colSums(contingency_table)

    test_result <- tryCatch({
      prop.trend.test(successes, totals)
    }, error = function(e) { return(NULL) })
  }
}

```

```

    if (!is.null(test_result)) {
      print(test_result)
    } else {
      cat("Error: Could not run Trend Test. Verify your variable setup.\n")
    }
  }

# --- 2. IF NUMERIC -> RUN T-TEST ---
} else if (is.numeric(d[[col_name]])) {
  cat("Predictor Type: NUMERIC -> Running Independent t-test...\n\n")

  test_result <- tryCatch({
    t.test(d[[col_name]] ~ d[[target_var]])
  }, error = function(e) { return(NULL) })

  if (!is.null(test_result)) {
    print(test_result)
  } else {
    cat("Error: Could not run t-test. Check for missing values.\n")
  }
}

# --- 3. IF NOMINAL / BINARY -> RUN CHI-SQUARE ---
} else if (is.factor(d[[col_name]]) || is.character(d[[col_name]])) {
  cat("Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...\n\n")

  contingency_table <- table(d[[target_var]], d[[col_name]])
  print(contingency_table)
  cat("\n")

  test_result <- tryCatch({
    chisq.test(contingency_table)
  }, error = function(e) { return(NULL) })

  if (!is.null(test_result)) {
    print(test_result)
  } else {
    cat("Error: Could not run Chi-Square test.\n")
  }
}
}
}

```

```

##
## =====
## Testing Relationship: mental_issue vs country_name
## =====
## Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...
##
##
##      Albania Armenia Austria Belgium (Flemish) Belgium (French) Canada Croatia
## 0      790      1361      2296                2247                1513      4124      2266
## 1      266       589       627                589                694      1467       668
##
##      Czech Republic England Estonia Finland France Georgia Germany Hungary
## 0              5744       159      2409      1769      3526      1660      2328      2010

```

```

##      1      2133      95      1061      582      1530      467      496      769
##
##      Iceland Ireland Israel Italy Kazakhstan Lithuania Luxembourg Malta
##      0      3309      432      2582      1739      1850      2059      1564      404
##      1      1121      171      1445      1268      329      662      607      208
##
##      Netherlands Norway Portugal Republic of Moldova Romania Russia Scotland
##      0      2491      1211      3750      2388      1681      2012      787
##      1      567      261      1205      842      819      592      281
##
##      Serbia Slovenia Spain Sweden Turkey Ukraine Wales
##      0      1984      3087      2415      1581      1640      2915      2015
##      1      618      887      549      772      1826      1426      817
##
##
##      Pearson's Chi-squared test
##
##      data: contingency_table
##      X-squared = 2776.1, df = 36, p-value < 2.2e-16
##
##
##      =====
##      Testing Relationship: mental_issue vs bodyheight
##      =====
##      Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##      Welch Two Sample t-test
##
##      data: d[[col_name]] by d[[target_var]]
##      t = -7.5384, df = 57278, p-value = 4.828e-14
##      alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
##      95 percent confidence interval:
##      -0.7394680 -0.4342877
##      sample estimates:
##      mean in group 0 mean in group 1
##      161.7640      162.3509
##
##
##      =====
##      Testing Relationship: mental_issue vs bodyweight
##      =====
##      Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##      Welch Two Sample t-test
##
##      data: d[[col_name]] by d[[target_var]]
##      t = -20.023, df = 53178, p-value < 2.2e-16
##      alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
##      95 percent confidence interval:
##      -2.017091 -1.657400
##      sample estimates:
##      mean in group 0 mean in group 1

```

```

##          51.55489          53.39214
##
##
## =====
## Testing Relationship: mental_issue vs MBMI
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -24.228, df = 49184, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -0.6638405 -0.5644728
## sample estimates:
## mean in group 0 mean in group 1
##      19.45514      20.06930
##
##
## =====
## Testing Relationship: mental_issue vs IOTF4
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3      4
## 0 10993 55631 9539 1935
## 1  4051 20106 4076 1073
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3 4
## X-squared = 105.21, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs agecat
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 23818 27346 26934
## 1  6269 10137 12900
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3

```

```

## X-squared = 1151.5, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs sex
## =====
## Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...
##
##
##      1      2
## 0 40500 37598
## 1 10256 19050
##
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: contingency_table
## X-squared = 2430.1, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs IRRELFAS_LMH
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 13352 49163 15583
## 1  5728 18149  5429
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3
## X-squared = 85.88, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs IRFAS
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = 19.655, df = 52226, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  0.3289788 0.4018589
## sample estimates:
## mean in group 0 mean in group 1
##      8.082576      7.717157

```

```

##
##
## =====
## Testing Relationship: mental_issue vs lifesat_low
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
## 0 73232 4866
## 1 21294 8012

##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 8997.7, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs mental_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -518.82, df = 36694, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -2.490436 -2.471690
## sample estimates:
## mean in group 0 mean in group 1
##      0.2641425      2.7452058
##
##
## =====
## Testing Relationship: mental_issue vs pmsu_lmh
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 49970 24610 3518
## 1 11592 14012 3702
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3
## X-squared = 5875.6, df = 1, p-value < 2.2e-16

```

```

##
##
## =====
## Testing Relationship: mental_issue vs pmsu_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -75.819, df = 44860, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -1.176923 -1.117607
## sample estimates:
## mean in group 0 mean in group 1
## 1.474109 2.621374
##
##
## =====
## Testing Relationship: mental_issue vs cbullied
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
## 0 1
## 0 71095 7003
## 1 23089 6217
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 2961.3, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs bulliedsum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -57.626, df = 38594, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -0.6125673 -0.5722677
## sample estimates:
## mean in group 0 mean in group 1
## 2.453955 3.046373

```

```

##
##
## =====
## Testing Relationship: mental_issue vs onlfreq_is
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
## 0 52129 25969
## 1 16484 12822

##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 1018.3, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs onlfreq_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -21.652, df = 51787, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -0.6664252 -0.5557844
## sample estimates:
## mean in group 0 mean in group 1
##      13.95915      14.57026
##
##
## =====
## Testing Relationship: mental_issue vs onlpref
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 44213 22098 11787
## 1 13274  8743  7289
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3
## X-squared = 1621.5, df = 1, p-value < 2.2e-16

```

```

##
##
## =====
## Testing Relationship: mental_issue vs onlpref_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -40.243, df = 48583, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -1.0689588 -0.9696696
## sample estimates:
## mean in group 0 mean in group 1
##      6.609260      7.628574
##
##
## =====
## Testing Relationship: mental_issue vs famsup_hml
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 5903 6733 65462
## 1 3837 6165 19304
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3
## X-squared = 2998.4, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs famsup_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = 61.311, df = 47945, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  2.803933 2.989129
## sample estimates:
## mean in group 0 mean in group 1
##      23.86316      20.96663

```

```

##
##
## =====
## Testing Relationship: mental_issue vs frisup_hml
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      1      2      3
## 0 7241 13510 57347
## 1 4175 6915 18216
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2 3
## X-squared = 1241.3, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs frisup_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = 35.397, df = 49170, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## 1.581894 1.767350
## sample estimates:
## mean in group 0 mean in group 1
##      22.04383      20.36921
##
##
## =====
## Testing Relationship: mental_issue vs teachersup
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
## 0 32592 45506
## 1 17828 11478
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 3121.8, df = 1, p-value < 2.2e-16

```

```

##
##
## =====
## Testing Relationship: mental_issue vs teachersup_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = 65.45, df = 46520, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## 1.247438 1.324457
## sample estimates:
## mean in group 0 mean in group 1
## 11.71313 10.42718
##
##
## =====
## Testing Relationship: mental_issue vs studsup
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
## 0 1
## 0 29098 49000
## 1 16868 12438
##
##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 3587, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs studsup_sum
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = 68.533, df = 45678, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## 1.206009 1.277023
## sample estimates:
## mean in group 0 mean in group 1
## 11.89548 10.65396

```

```

##
##
## =====
## Testing Relationship: mental_issue vs talkf
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
##  0 21401 56697
##  1 14959 14347

##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 5318.8, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs talkm
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
##  0  9296 68802
##  1  8470 20836

##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 4460.4, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs talkscore
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -78.768, df = 46097, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -1.0182352 -0.9687915
## sample estimates:
## mean in group 0 mean in group 1
##      3.732311      4.725824

```

```

##
##
## =====
## Testing Relationship: mental_issue vs timeexe_r
## =====
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##      0      1
## 0 19618 58480
## 1 10393 18913

##
## Chi-squared Test for Trend in Proportions
##
## data: successes out of totals ,
## using scores: 1 2
## X-squared = 1132.4, df = 1, p-value < 2.2e-16
##
##
## =====
## Testing Relationship: mental_issue vs timeexe
## =====
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
## Welch Two Sample t-test
##
## data: d[[col_name]] by d[[target_var]]
## t = -33.901, df = 47234, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -0.4099063 -0.3650993
## sample estimates:
## mean in group 0 mean in group 1
##      2.900971      3.288473

# # chi for nominal
# cont_tbl <- table(d[["mental_issue"]], d[["pmsu_lmh"]])
# print(cont_tbl)
# chi <- chisq.test(cont_tbl)
# print(chi)
# print(chi$p.value)
# print(chi$statistic)
#
# # cochran for ordered
# cont_tbl <- table(d[["mental_issue"]], d[["pmsu_lmh"]])
# cont_tbl
# successes <- cont_tbl["1", ]
# successes
# totals <- colSums(cont_tbl)
# coch <- prop.trend.test(successes, totals)
# coch
#

```

```

# # t.test for num
# tt <- t.test(d[["pmsu_sum"]] ~ d[["mental_issue"]])
# tt

# --- 1. SETUP YOUR TARGET VARIABLE NAME ---
# Target variable is set to "mental_issue" and dataset is named "d"
target_var <- "mental_issue"

# Ensure target is a factor and identify the "success" level (the 2nd level)
d[[target_var]] <- as.factor(d[[target_var]])
target_levels <- levels(d[[target_var]])
success_level <- target_levels[2] # Grabs group 1 / "Yes" status automatically

# --- 2. INITIALIZE AN EMPTY DATAFRAME FOR THE SUMMARY TABLE ---
summary_table <- data.frame(
  Predictor_Variable = character(),
  Variable_Type      = character(),
  Statistical_Test    = character(),
  Test_Statistic      = character(),
  p_value            = character(),
  stringsAsFactors   = FALSE
)

# --- 3. THE AUTOMATED LOOP ---
for (col_name in names(d)) {

  # Skip the target variable itself
  if (col_name == target_var) next

  # Create baseline variables for storing this row's data
  var_type <- "Unknown"
  test_name <- "None"
  stat_val <- "N/A"
  p_val_str <- "N/A"

  # --- 1. IF ORDINAL -> RUN COCHRAN-ARMITAGE TREND TEST ---
  if (is.ordered(d[[col_name]])) {
    var_type <- "Ordinal"
    test_name <- "Cochran-Armitage Trend Test"

    contingency_table <- table(d[[target_var]], d[[col_name]])
    successes <- contingency_table[success_level, ]
    totals <- colSums(contingency_table)

    test_result <- tryCatch({ prop.trend.test(successes, totals) }, error = function(e) { return(NULL) })

    if (!is.null(test_result)) {
      stat_val <- paste0("X-squared = ", round(test_result$statistic, 2))
      p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
    }

    # --- 2. IF NUMERIC -> RUN T-TEST ---
  } else if (is.numeric(d[[col_name]])) {

```

```

var_type <- "Numeric"
test_name <- "Welch Two Sample t-test"

test_result <- tryCatch({ t.test(d[[col_name]] ~ d[[target_var]]) }, error = function(e) { return(NULL) })

if (!is.null(test_result)) {
  stat_val <- paste0("t(", round(test_result$parameter, 1), ") = ", round(test_result$statistic, 2))
  p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
}

# --- 3. IF NOMINAL / BINARY -> RUN CHI-SQUARE ---
} else if (is.factor(d[[col_name]]) || is.character(d[[col_name]])) {
  var_type <- if(length(unique(d[[col_name]])) == 2) "Binary" else "Nominal"
  test_name <- "Pearson's Chi-Square Test"

  contingency_table <- table(d[[target_var]], d[[col_name]])
  test_result <- tryCatch({ chisq.test(contingency_table) }, error = function(e) { return(NULL) })

  if (!is.null(test_result)) {
    stat_val <- paste0("X-squared(", test_result$parameter, ") = ", round(test_result$statistic, 2))
    p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
  }
}

# --- 4. APPEND THE RESULT ROW TO OUR TABLE ---
new_row <- data.frame(
  Predictor_Variable = col_name,
  Variable_Type      = var_type,
  Statistical_Test    = test_name,
  Test_Statistic      = stat_val,
  p_value            = p_val_str,
  stringsAsFactors   = FALSE
)
summary_table <- rbind(summary_table, new_row)
}

# --- 4. VIEW THE FINAL SUMMARY TABLE ---
kable(summary_table)

```

Predictor_Variable	Variable_Type	Statistical_Test	Test_Statistic	p_value
seqno_int	Numeric	Welch Two Sample t-test	t(51650) = -12.6	< 0.001
country_name	Nominal	Pearson's Chi-Square Test	X-squared(36) = 2776.08	< 0.001
bodyheight	Numeric	Welch Two Sample t-test	t(57278.4) = -7.54	< 0.001
bodyweight	Numeric	Welch Two Sample t-test	t(53178.4) = -20.02	< 0.001
MBMI	Numeric	Welch Two Sample t-test	t(49183.8) = -24.23	< 0.001
IOTF4	Ordinal	Cochran-Armitage Trend Test	X-squared = 105.21	< 0.001
agecat	Ordinal	Cochran-Armitage Trend Test	X-squared = 1151.5	< 0.001
sex	Binary	Pearson's Chi-Square Test	X-squared(1) = 2430.11	< 0.001
IRRELFAS_LMH	Ordinal	Cochran-Armitage Trend Test	X-squared = 85.88	< 0.001

Predictor_Variable	Variable_Type	Statistical_Test	Test_Statistic	p_value
IRFAS	Numeric	Welch Two Sample t-test	t(52225.8) = 19.65	< 0.001
lifesat_low	Ordinal	Cochran-Armitage Trend Test	X-squared = 8997.69	< 0.001
mental_sum	Numeric	Welch Two Sample t-test	t(36694.3) = -518.82	< 0.001
pmsu_lmh	Ordinal	Cochran-Armitage Trend Test	X-squared = 5875.59	< 0.001
pmsu_sum	Numeric	Welch Two Sample t-test	t(44860.2) = -75.82	< 0.001
cbullied	Ordinal	Cochran-Armitage Trend Test	X-squared = 2961.27	< 0.001
bulliedsum	Numeric	Welch Two Sample t-test	t(38593.8) = -57.63	< 0.001
onlfreq_is	Ordinal	Cochran-Armitage Trend Test	X-squared = 1018.32	< 0.001
onlfreq_sum	Numeric	Welch Two Sample t-test	t(51786.7) = -21.65	< 0.001
onlpref	Ordinal	Cochran-Armitage Trend Test	X-squared = 1621.46	< 0.001
onlpref_sum	Numeric	Welch Two Sample t-test	t(48582.5) = -40.24	< 0.001
famsup_hml	Ordinal	Cochran-Armitage Trend Test	X-squared = 2998.45	< 0.001
famsup_sum	Numeric	Welch Two Sample t-test	t(47945.3) = 61.31	< 0.001
frisup_hml	Ordinal	Cochran-Armitage Trend Test	X-squared = 1241.26	< 0.001
frisup_sum	Numeric	Welch Two Sample t-test	t(49170) = 35.4	< 0.001
teachersup	Ordinal	Cochran-Armitage Trend Test	X-squared = 3121.83	< 0.001
teachersup_sum	Numeric	Welch Two Sample t-test	t(46519.5) = 65.45	< 0.001
studsup	Ordinal	Cochran-Armitage Trend Test	X-squared = 3586.98	< 0.001
studsup_sum	Numeric	Welch Two Sample t-test	t(45678.3) = 68.53	< 0.001
talkf	Ordinal	Cochran-Armitage Trend Test	X-squared = 5318.78	< 0.001
talkm	Ordinal	Cochran-Armitage Trend Test	X-squared = 4460.45	< 0.001
talkscore	Numeric	Welch Two Sample t-test	t(46097) = -78.77	< 0.001
timeexe_r	Ordinal	Cochran-Armitage Trend Test	X-squared = 1132.43	< 0.001
timeexe	Numeric	Welch Two Sample t-test	t(47234.5) = -33.9	< 0.001

```

## create modeling dat
# dat <- dat %>%
#   mutate(y = ifelse(lifesat >5,1,0),
#          y = factor(y))

d_md1 <- d %>%
  select(country_name, MBMI, agecat, sex, IRRELFAS_LMH, mental_issue, pmsu_lmh, cbullied, onlfreq_is, onlpref, famsup_hml, frisup_hml, teachersup, studsup, talkf, talkm, talkscore, timeexe_r, timeexe)

#baseline
d_md1 %>%
  group_by(mental_issue) %>%
  tally() %>%
  mutate(prop = n/sum(n))

## # A tibble: 2 x 3

```

```
##   mental_issue      n  prop
##   <ord>           <int> <dbl>
## 1 0               78098 0.727
## 2 1               29306 0.273

# data split
set.seed(67)
data_split <- initial_split(d_mdl, prop = 0.8, strata = mental_issue)
train_data <- training(data_split)
test_data  <- testing(data_split)

# 3. Create a Preprocessing Recipe -----
# Define formula (Outcome ~ Predictors) and add preprocessing steps
lr_recipe <- recipe(mental_issue ~ . ,
                    data = train_data) %>%
  step_rm(country_name)
# step_zv(all_predictors()) %>%           # Remove zero-variance predictors
# step_normalize(all_numeric_predictors()) # Center and scale numeric variables

# 4. Define the Logistic Regression Model Spec -----
lr_spec <- logistic_reg() %>%
  set_engine("glm") %>%           # Use standard generalized linear model
  set_mode("classification")     # Specify binary classification mode

# 5. Create the Workflow -----
# Bundle the recipe and model specification together
lr_workflow <- workflow() %>%
  add_recipe(lr_recipe) %>%
  add_model(lr_spec)

# 6. Fit the Model (Train) -----
# This automatically preps the recipe and trains the model on train data
lr_fit <- fit(lr_workflow, data = train_data)

# 7. Evaluate on Test Data -----
# Predict classes (labels)
predictions_class <- predict(lr_fit, new_data = test_data)
predictions_class

## # A tibble: 21,482 x 1
##   .pred_class
##   <ord>
## 1 0
## 2 0
## 3 0
## 4 0
## 5 0
## 6 0
## 7 0
## 8 0
## 9 0
## 10 0
## # i 21,472 more rows
```

```
# Predict probabilities
predictions_prob <- predict(lr_fit, new_data = test_data, type = "prob")
predictions_prob
```

```
## # A tibble: 21,482 x 2
##   .pred_0 .pred_1
##   <dbl>   <dbl>
## 1  0.768  0.232
## 2  0.826  0.174
## 3  0.902  0.0976
## 4  0.765  0.235
## 5  0.943  0.0566
## 6  0.873  0.127
## 7  0.884  0.116
## 8  0.846  0.154
## 9  0.902  0.0984
## 10 0.834  0.166
## # i 21,472 more rows
```

```
# Combine test data with predictions
evaluation_data <- test_data %>%
  bind_cols(predictions_class, predictions_prob)

# Print confusion matrix and accuracy metrics
evaluation_data %>% conf_mat(truth = mental_issue, estimate = .pred_class)
```

```
##           Truth
## Prediction    0    1
##           0 14615 4092
##           1  1005 1770
```

```
evaluation_data %>% metrics(truth = mental_issue, estimate = .pred_class)
```

```
## # A tibble: 2 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 accuracy binary      0.763
## 2 kap     binary      0.284
```

```
# Extract and print clean model coefficients
lr_fit %>%
  extract_fit_parsnip() %>%
  tidy() %>%
  print(n = Inf)
```

```
## # A tibble: 22 x 5
##   term          estimate std.error statistic  p.value
##   <chr>         <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)  -0.708    0.0528   -13.4  4.53e- 41
## 2 MBMI         0.0180    0.00242    7.41  1.24e- 13
## 3 agecat.L     0.0937    0.0166    5.66  1.51e- 8
```

```
## 4 agecat.Q      0.0372    0.0146      2.56 1.05e- 2
## 5 sex2          0.550     0.0175     31.4 3.20e-216
## 6 IRRELFAS_LMH.L 0.00201  0.0196      0.103 9.18e- 1
## 7 IRRELFAS_LMH.Q 0.00257  0.0141      0.182 8.56e- 1
## 8 pmsu_lmh.L    0.652     0.0226     28.9 1.20e-183
## 9 pmsu_lmh.Q   -0.0701   0.0164     -4.28 1.87e- 5
## 10 cbullied.L   0.400     0.0168     23.9 8.01e-126
## 11 onlfreq_is.L 0.201     0.0124     16.2 1.00e- 58
## 12 onlpref.L    0.170     0.0161     10.6 4.00e- 26
## 13 onlpref.Q    0.117     0.0154      7.59 3.20e- 14
## 14 famsup_hml.L -0.222     0.0209    -10.6 2.36e- 26
## 15 famsup_hml.Q -0.206     0.0215     -9.56 1.21e- 21
## 16 frisup_hml.L -0.214     0.0196    -10.9 9.49e- 28
## 17 frisup_hml.Q -0.0487   0.0186     -2.61 9.00e- 3
## 18 teachersup.L -0.234     0.0127    -18.4 1.33e- 75
## 19 studsup.L    -0.299     0.0126    -23.7 1.16e-124
## 20 talkf.L      -0.287     0.0132    -21.8 4.59e-105
## 21 talkm.L      -0.311     0.0160    -19.4 1.02e- 83
## 22 timeexe_r.L  -0.155     0.0130    -12.0 5.99e- 33
```

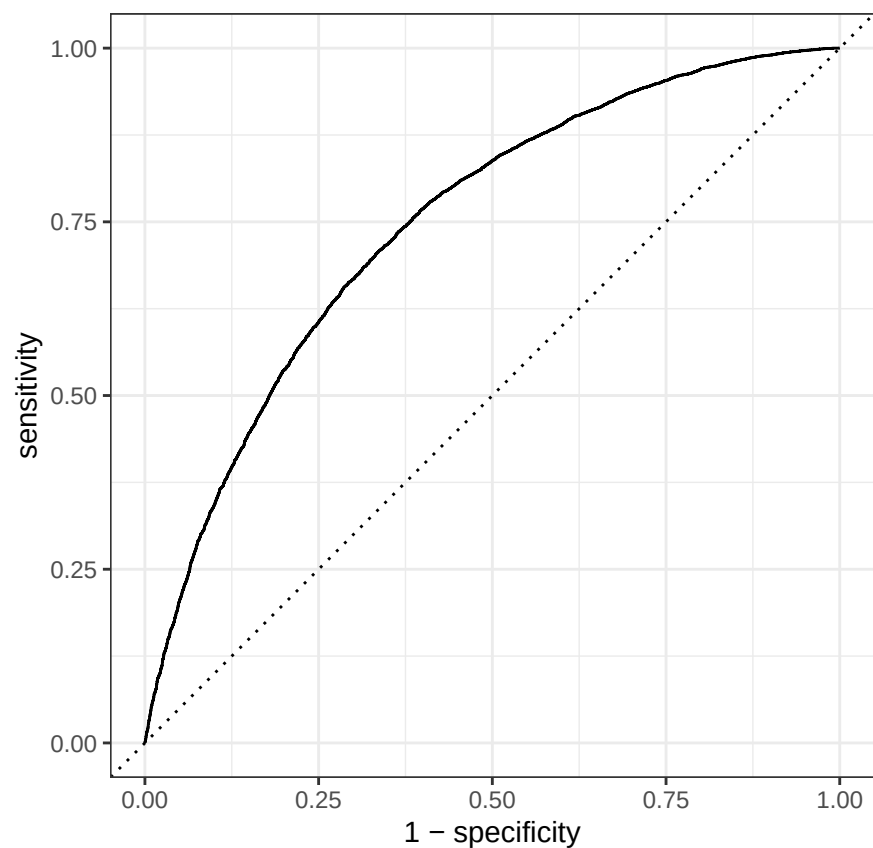
```
# Print coefficients as Odds Ratios
lr_fit %>%
  extract_fit_parsnip() %>%
  tidy(exponentiate = TRUE) %>%
  mutate(is_sig = ifelse(p.value < .05, 1, 0)) %>%
  print(n = Inf)
```

```
## # A tibble: 22 x 6
##   term      estimate std.error statistic  p.value is_sig
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
## 1 (Intercept)  0.493    0.0528   -13.4  4.53e- 41      1
## 2 MBMI        1.02    0.00242    7.41  1.24e- 13      1
## 3 agecat.L     1.10    0.0166     5.66  1.51e- 8       1
## 4 agecat.Q     1.04    0.0146     2.56  1.05e- 2       1
## 5 sex2        1.73    0.0175    31.4  3.20e-216      1
## 6 IRRELFAS_LMH.L 1.00    0.0196     0.103 9.18e- 1       0
## 7 IRRELFAS_LMH.Q 1.00    0.0141     0.182 8.56e- 1       0
## 8 pmsu_lmh.L   1.92    0.0226     28.9  1.20e-183      1
## 9 pmsu_lmh.Q   0.932   0.0164     -4.28  1.87e- 5       1
## 10 cbullied.L  1.49    0.0168     23.9  8.01e-126      1
## 11 onlfreq_is.L 1.22    0.0124     16.2  1.00e- 58      1
## 12 onlpref.L   1.19    0.0161     10.6  4.00e- 26      1
## 13 onlpref.Q   1.12    0.0154      7.59  3.20e- 14      1
## 14 famsup_hml.L 0.801   0.0209    -10.6  2.36e- 26      1
## 15 famsup_hml.Q 0.814   0.0215     -9.56  1.21e- 21      1
## 16 frisup_hml.L 0.807   0.0196    -10.9  9.49e- 28      1
## 17 frisup_hml.Q 0.952   0.0186     -2.61  9.00e- 3       1
## 18 teachersup.L 0.791   0.0127    -18.4  1.33e- 75      1
## 19 studsup.L   0.742   0.0126    -23.7  1.16e-124      1
## 20 talkf.L     0.751   0.0132    -21.8  4.59e-105      1
## 21 talkm.L     0.733   0.0160    -19.4  1.02e- 83      1
## 22 timeexe_r.L 0.856   0.0130    -12.0  5.99e- 33      1
```

```
# Calculate ROC AUC
evaluation_data %>%
  roc_auc(truth = mental_issue, .pred_0)
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 roc_auc binary      0.748
```

```
# Generate and plot the ROC Curve
evaluation_data %>%
  roc_curve(truth = mental_issue, .pred_0) %>%
  autoplot()
```



```
# 1. Isolate just your predictor columns (exclude your target "mental_issue")
# Hint: Use a subset of up to 1,000-2,000 rows for speed
X_explain <- d[1:1000, ] %>% select(-mental_issue)

require(kernelshap)
require(shapviz)

# 2. Compute the SHAP values
# Type = "prob" ensures R looks at the predicted probabilities (e.g., .pred_Yes)
shap_results <- kernelshap(
```

```

object = lr_fit,
X = X_explain,
bg_X = d[1:200, ] %>% select(-mental_issue), # Small background baseline
type = "prob"
)

```

```
## |
```

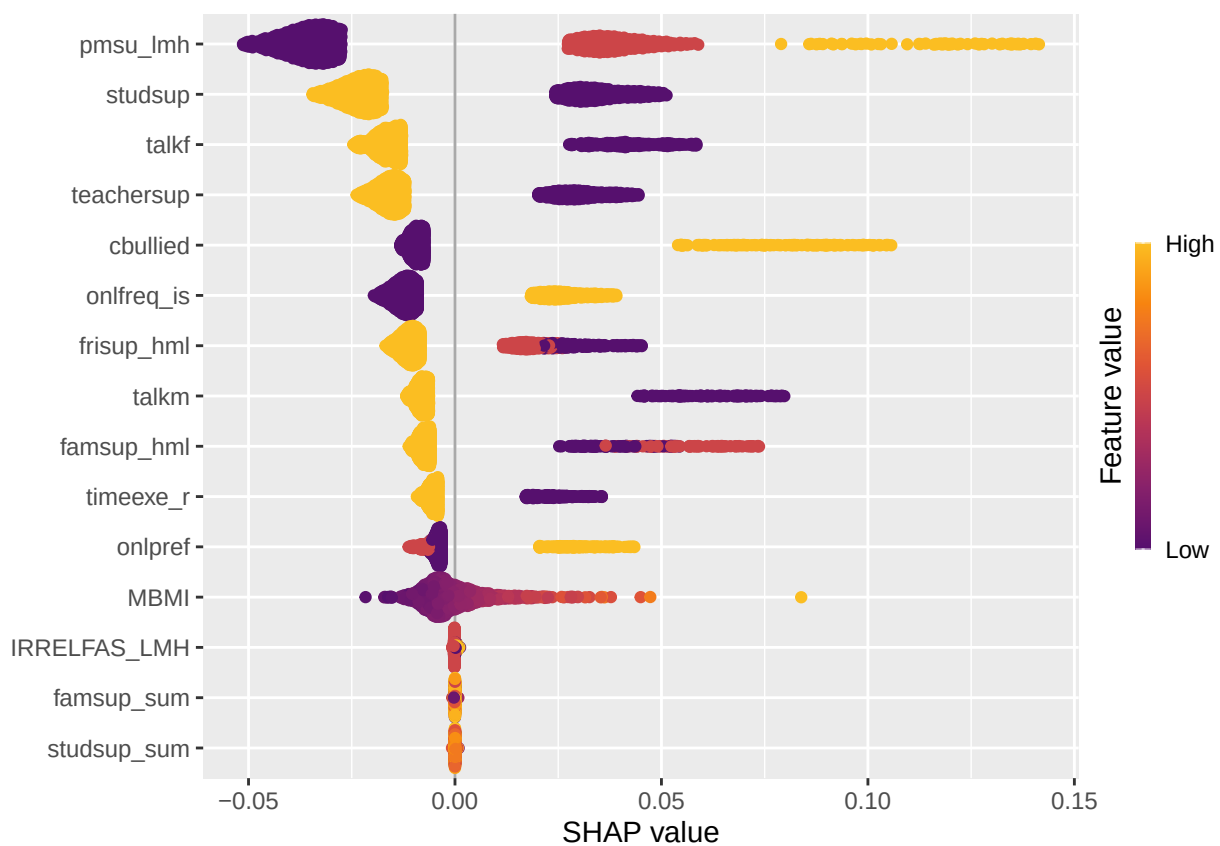
```
|
```

```

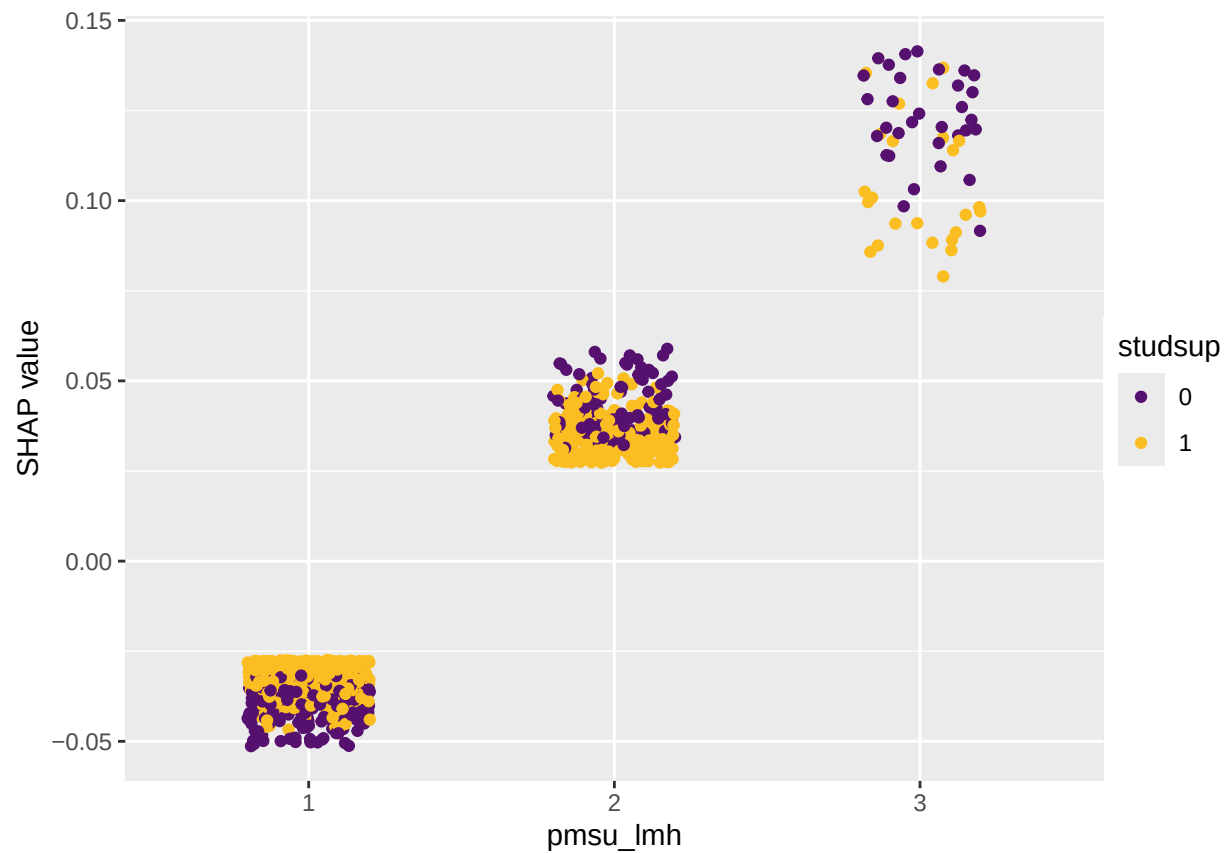
# 3. Convert to a shapviz object for instant plotting
shp_viz <- shapviz(shap_results)

# Replace ".pred_1" with your positive class prediction column name if different
sv_importance(shp_viz$.pred_1, kind = "beeswarm")

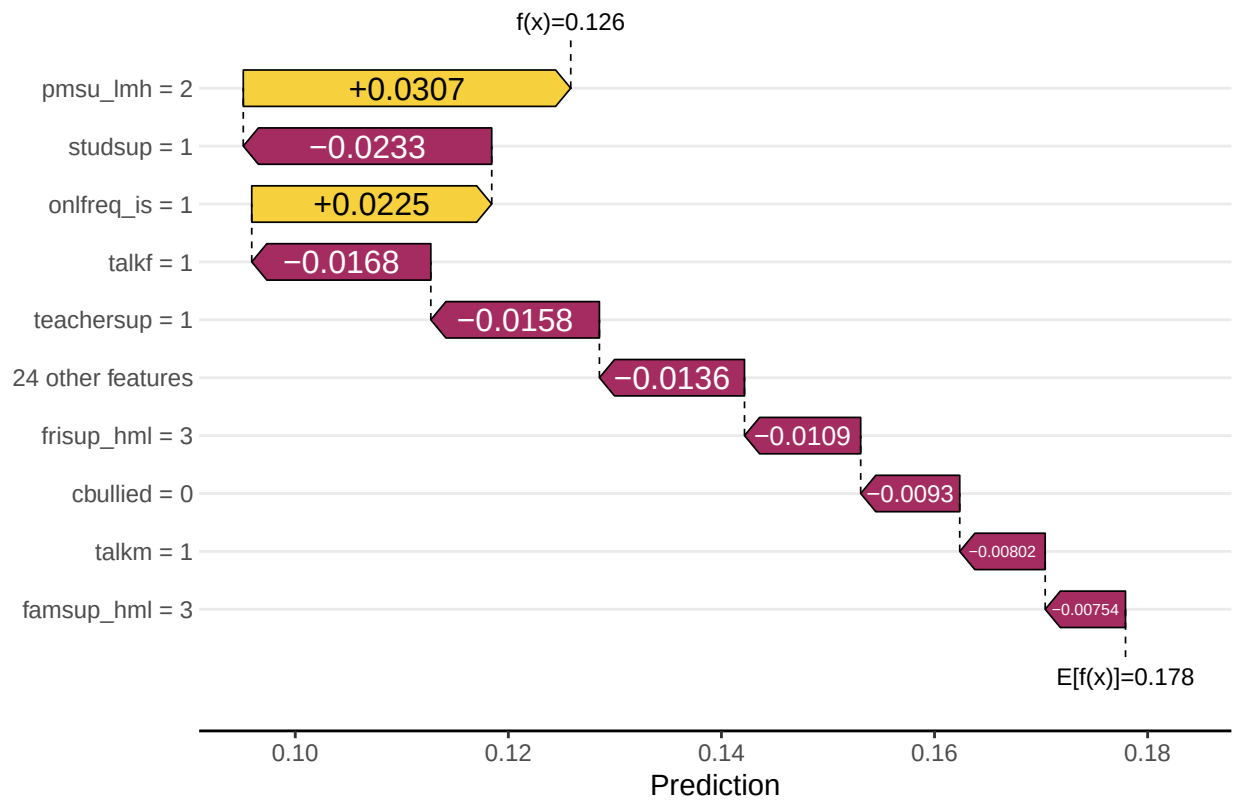
```

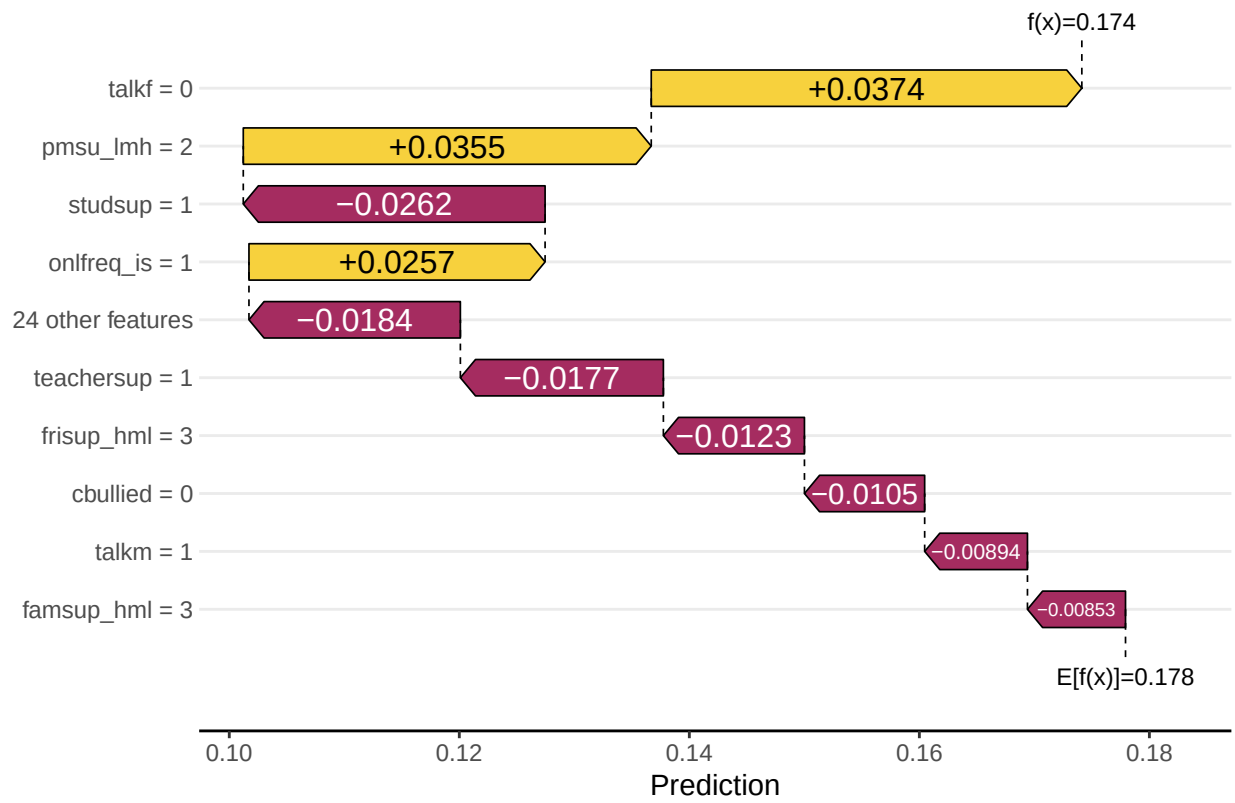


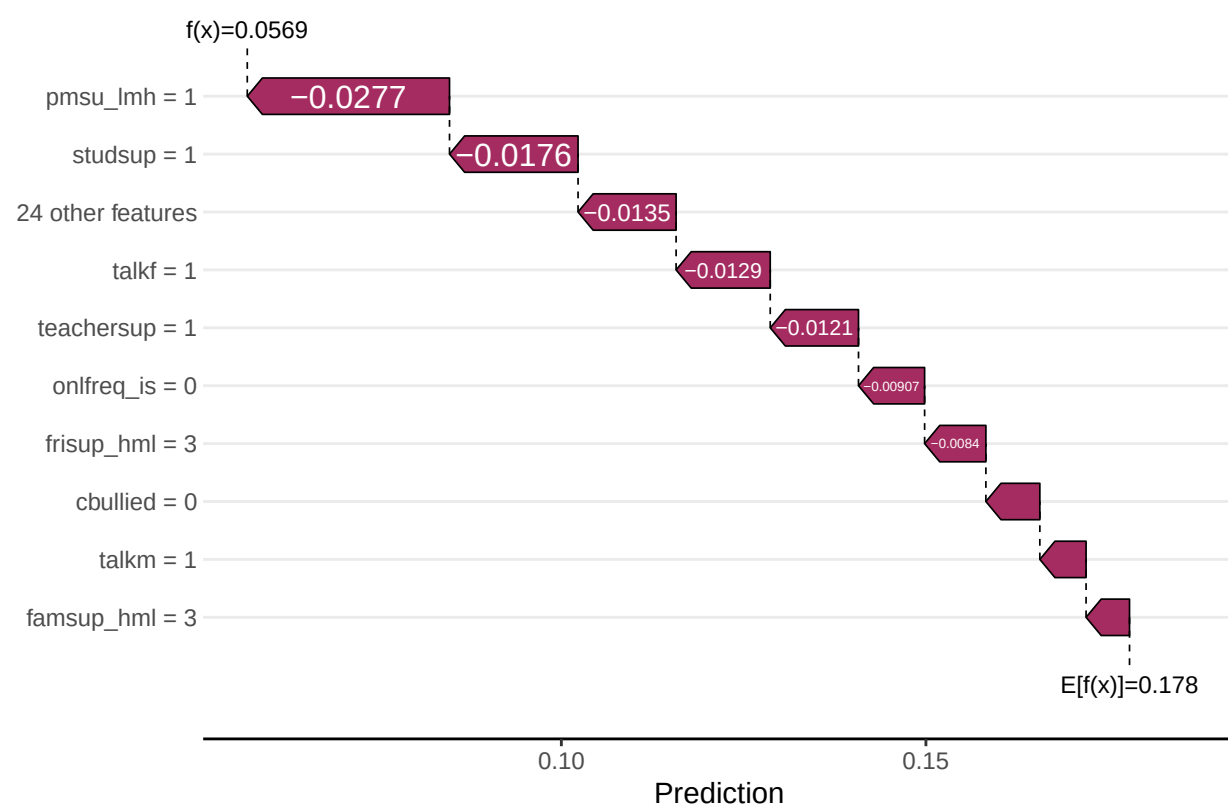
```
sv_dependence(shp_viz$.pred_1, v = "pmsu_lmh")
```



```
for (i in c(1,3,5)){  
  print(sv_waterfall(shp_viz$.pred_1, row_id = i))  
}
```







```
#sv_waterfall(shp_viz$.pred_1, row_id = 5)
```